Clean statement specimens

Prototype 0.1

These are statement renderings, not proof certificates. The collection specializes Mathlib results for syntax comparison. The algorithms section contains CLR-aligned sorting results and number-theoretic supporting lemmas, not a formalization of CLR.

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1 Stewart example

Result 1 (similarity transforms eigenvector). [Stewart 2001, Theorem 1.10, page 9]
Let n be a natural number. Let A and S be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Let μ be a complex number. Assume $\det(S) \neq 0$ (`invertibility`). Assume $x \neq 0$ (`nonzero`). Assume $A\,x = \mu\,x$ (`eigenpair`). Let $y = S^{-1}\,x$. Then

$$y \neq 0 \text{ and } S^{-1}\,A\,S\,y = \mu\,y.$$

Adapted from the cited source. Scope: eigenvector part.

2 Linear algebra

Result 2 (determinant product). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B) = \det(A)\,\det(B).$$

Result 3 (determinant transpose). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\det(A^{\mathsf{T}}) = \det(A).$$

Result 4 (determinant product swap). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B) = \det(B\,A).$$

Result 5 (determinant left swap). [Mathlib]
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,(B\,C)) = \det(B\,(A\,C)).$$

Result 6 (determinant right swap). [Mathlib]
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B\,C) = \det(A\,C\,B).$$

Result 7 (determinant power). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let k be a natural number. Then

$$\det(A^k) = \det(A)^k.$$

Result 8 (trace sum). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B).$$

Result 9 (trace difference). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}(A - B) = \operatorname{tr}(A) - \operatorname{tr}(B).$$

Result 10 (trace negative). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\operatorname{tr}(-A) = -\operatorname{tr}(A).$$

Result 11 (trace transpose). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\operatorname{tr}(A^{\mathsf{T}}) = \operatorname{tr}(A).$$

Result 12 (trace product swap). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}(A\,B) = \operatorname{tr}(B\,A).$$

Result 13 (trace cycle). [Mathlib]
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}(A\,B\,C) = \operatorname{tr}(C\,A\,B).$$

Result 14 (matrix vector associativity). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Then

$$A\,(B\,x) = A\,B\,x.$$

Result 15 (matrix vector addition). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let x and y be n-vectors over $\mathbb{C}$. Then

$$A\,(x + y) = A\,x + A\,y.$$

Result 16 (matrix addition action). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Then

$$(A + B)\,x = A\,x + B\,x.$$

Result 17 (matrix vector subtraction). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let x and y be n-vectors over $\mathbb{C}$. Then

$$A\,(x - y) = A\,x - A\,y.$$

Result 18 (matrix subtraction action). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Then

$$(A - B)\,x = A\,x - B\,x.$$

Result 19 (matrix scalar action). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Let c be a complex number. Then

$$A\,(c\,x) = c\,(A\,x).$$

Result 20 (transpose involution). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$A^{\mathsf{T}\mathsf{T}} = A.$$

Result 21 (transpose product). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$A\,B^{\mathsf{T}} = B^{\mathsf{T}}\,A^{\mathsf{T}}.$$

Result 22 (trace commutator). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\operatorname{tr}([A, B]) = 0.$$

Adapted from the cited source.

Result 23 (trace matrix square). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\operatorname{tr}((A)^2) = \operatorname{tr}(A\,A).$$

Adapted from the cited source.

Result 24 (determinant matrix square). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\det((A)^2) = \det(A)\,\det(A).$$

Adapted from the cited source.

3 Graph theory

Result 25 (adjacency symmetry). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Assume u is adjacent to v in G (`edge`). Then

$$v \text{ is adjacent to } u \text{ in } G.$$

Result 26 (adjacency commutes). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } G \text{ if and only if } v \text{ is adjacent to } u \text{ in } G.$$

Result 27 (no self edges). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$\text{not } (u \text{ is adjacent to } u \text{ in } G).$$

Result 28 (adjacent vertices distinct). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Assume u is adjacent to v in G (`edge`). Then

$$u \neq v.$$

Result 29 (union adjacency). [Mathlib]
Let n be a natural number. Let G and H be simple graphs on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } (G \cup H) \text{ if and only if } u \text{ is adjacent to } v \text{ in } G \text{ or } u \text{ is adjacent to } v \text{ in } H.$$

Result 30 (intersection adjacency). [Mathlib]
Let n be a natural number. Let G and H be simple graphs on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } (G \cap H) \text{ if and only if } u \text{ is adjacent to } v \text{ in } G \text{ and } u \text{ is adjacent to } v \text{ in } H.$$

Result 31 (complement adjacency). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } \overline{G} \text{ if and only if } u \neq v \text{ and not } (u \text{ is adjacent to } v \text{ in } G).$$

Result 32 (support monotonicity). [Mathlib]
Let n be a natural number. Let G and H be simple graphs on $\{0, \ldots, n-1\}$. Assume G is a subgraph of H (`inclusion`). Then

$$\text{supp}(G) \text{ is a subset of } \text{supp}(H).$$

Result 33 (neighbors in support). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \text{ is a subset of } \operatorname{supp}(G).$$

Result 34 (neighbor membership). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$v \text{ belongs to } N_G(u) \text{ if and only if } u \text{ is adjacent to } v \text{ in } G.$$

Result 35 (not own neighbor). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$\text{not } (u \text{ belongs to } N_G(u)).$$

Result 36 (common neighbors intersection). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \cap N_G(v) = N_G(u) \cap N_G(v).$$

Result 37 (common neighbors membership). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u, v and w be vertices in $\{0, \ldots, n-1\}$. Then

$$w \text{ belongs to } N_G(u) \cap N_G(v) \text{ if and only if } u \text{ is adjacent to } w \text{ in } G \text{ and } v \text{ is adjacent to } w \text{ in } G.$$

Result 38 (common neighbors symmetry). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \cap N_G(v) = N_G(v) \cap N_G(u).$$

Result 39 (common neighbors left). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \cap N_G(v) \text{ is a subset of } N_G(u).$$

Result 40 (common neighbors right). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$N_G(u) \cap N_G(v) \text{ is a subset of } N_G(v).$$

Result 41 (reachability reflexive). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$u \text{ is reachable from } u \text{ in } G.$$

Result 42 (reachability symmetric). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Assume v is reachable from u in G (`path`). Then

$$u \text{ is reachable from } v \text{ in } G.$$

Result 43 (reachability transitive). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u, v and w be vertices in $\{0, \ldots, n-1\}$. Assume v is reachable from u in G (`first`). Assume w is reachable from v in G (`second`). Then

$$w \text{ is reachable from } u \text{ in } G.$$

Result 44 (edge implies reachability). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Assume u is adjacent to v in G (`edge`). Then

$$v \text{ is reachable from } u \text{ in } G.$$

Result 45 (nonadjacency symmetry). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ and } v \text{ are nonadjacent in } G \text{ if and only if } v \text{ and } u \text{ are nonadjacent in } G.$$

Adapted from the cited source.

Result 46 (nonadjacency reflexive). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u be a vertex in $\{0, \ldots, n-1\}$. Then

$$u \text{ and } u \text{ are nonadjacent in } G.$$

Adapted from the cited source.

Result 47 (complement from nonadjacency). [Mathlib]
Let n be a natural number. Let G be a simple graph on $\{0, \ldots, n-1\}$. Let u and v be vertices in $\{0, \ldots, n-1\}$. Then

$$u \text{ is adjacent to } v \text{ in } \overline{G} \text{ if and only if } u \neq v \text{ and } u \text{ and } v \text{ are nonadjacent in } G.$$

Adapted from the cited source.

4 Analysis

Result 48 (exponential positive). [Mathlib]
Let x be a real number. Then

$$0 < \exp(x).$$

Result 49 (exponential nonzero). [Mathlib]
Let x be a real number. Then

$$\exp(x) \neq 0.$$

Result 50 (exponential sum). [Mathlib]
Let x and y be real numbers. Then

$$\exp(x + y) = \exp(x)\ \exp(y).$$

Result 51 (exponential difference). [Mathlib]
Let x and y be real numbers. Then

$$\exp(x - y) = \frac{\exp(x)}{\exp(y)}.$$

Result 52 (logarithm exponential). [Mathlib]
Let x be a real number. Then

$$\log(\exp(x)) = x.$$

Result 53 (exponential logarithm). [Mathlib]
Let x be a real number. Assume $0 < x$ (`positive`). Then

$$\exp(\log(x)) = x.$$

Result 54 (logarithm product). [Mathlib]
Let x and y be real numbers. Assume $x \neq 0$ (`nonzero_x`). Assume $y \neq 0$ (`nonzero_y`). Then

$$\log(x\,y) = \log(x) + \log(y).$$

Result 55 (logarithm quotient). [Mathlib]
Let x and y be real numbers. Assume $x \neq 0$ (`nonzero_x`). Assume $y \neq 0$ (`nonzero_y`). Then

$$\log(\frac{x}{y}) = \log(x) - \log(y).$$

Result 56 (square root nonnegative). [Mathlib]
Let x be a real number. Then

$$0 \leq \sqrt{x}.$$

Result 57 (square of square root). [Mathlib]
Let x be a real number. Assume $0 \leq x$ (`nonnegative`). Then

$$\sqrt{x}^2 = x.$$

Result 58 (square root of square). [Mathlib]
Let x be a real number. Then

$$\sqrt{x^2} = |x|.$$

Result 59 (square root product). [Mathlib]
Let x and y be real numbers. Assume $0 \leq x$ (`nonnegative`). Then

$$\sqrt{x\,y} = \sqrt{x}\,\sqrt{y}.$$

Result 60 (trigonometric identity). [Mathlib]
Let x be a real number. Then

$$\sin(x)^2 + \cos(x)^2 = 1.$$

Result 61 (sine addition). [Mathlib]
Let x and y be real numbers. Then

$$\sin(x + y) = \sin(x)\,\cos(y) + \cos(x)\,\sin(y).$$

Result 62 (cosine addition). [Mathlib]
Let x and y be real numbers. Then

$$\cos(x + y) = \cos(x)\,\cos(y) - \sin(x)\,\sin(y).$$

Result 63 (sine odd). [Mathlib]
Let x be a real number. Then

$$\sin(-x) = -\sin(x).$$

Result 64 (cosine even). [Mathlib]
Let x be a real number. Then

$$\cos(-x) = \cos(x).$$

Result 65 (continuous composition). [Mathlib]
Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$. Assume f is continuous (`hf`). Assume g is continuous (`hg`). Then

$$(f \circ g) \text{ is continuous.}$$

Result 66 (continuous sum). [Mathlib]
Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$. Assume f is continuous (`hf`). Assume g is continuous (`hg`). Then

$$f + g \text{ is continuous.}$$

Result 67 (continuous product). [Mathlib]
Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$. Assume f is continuous (`hf`). Assume g is continuous (`hg`). Then

$$f\ g \text{ is continuous.}$$

Result 68 (authored square root). [Mathlib]
Let x be a real number. Then

$$\sqrt{(x)^2} = |x|.$$

Adapted from the cited source.

Result 69 (authored trigonometric energy). [Mathlib]
Let x be a real number. Then

$$\text{trigonometric energy}(x) = 1.$$

Adapted from the cited source.

Result 70 (distance square root). [Mathlib]
Let x and y be real numbers. Then

$$\sqrt{\text{squared distance}(x, y)} = |x - y|.$$

Adapted from the cited source.

5 Undergraduate algorithms

Result 71 (insertion sort sorted). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{insertionSort}(xs) \text{ is sorted.}$$

Result 72 (insertion sort permutation). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{insertionSort}(xs) \text{ is a permutation of } xs.$$

Result 73 (insertion sort length). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$|\text{insertionSort}(xs)| = |xs|.$$

Result 74 (insertion sort membership). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Then

$$a \text{ belongs to } \text{insertionSort}(xs) \text{ if and only if } a \text{ belongs to } xs.$$

Result 75 (sorted input unchanged). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Assume xs is sorted (`ordered`). Then

$$\text{insertionSort}(xs) = xs.$$

Result 76 (ordered insert length). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Then

$$|\text{insert}(a, xs)| = |xs| + 1.$$

Result 77 (ordered insert membership). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a and b be natural numbers. Then

$$b \text{ belongs to } \text{insert}(a, xs) \text{ if and only if } b = a \text{ or } b \text{ belongs to } xs.$$

Result 78 (ordered insert permutation). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Then

$$\text{insert}(a, xs) \text{ is a permutation of } (a :: xs).$$

Result 79 (ordered insert preserves sortedness). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Assume xs is sorted (`ordered`). Then

$$\text{insert}(a, xs) \text{ is sorted.}$$

Result 80 (merge sort sorted). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{mergeSort}(xs) \text{ is sorted.}$$

Result 81 (merge sort sorted input). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Assume xs is sorted (`ordered`). Then

$$\text{mergeSort}(xs) = xs.$$

Result 82 (sorting algorithms agree). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{mergeSort}(xs) = \text{insertionSort}(xs).$$

Result 83 (merge preserves sortedness). [Mathlib]
Let xs and ys be lists with entries in $\mathbb{N}$. Assume xs is sorted (`hx`). Assume ys is sorted (`hy`). Then

$$\text{merge}(xs, ys) \text{ is sorted.}$$

Result 84 (insertion sort subsequence). [Mathlib]
Let xs and ys be lists with entries in $\mathbb{N}$. Assume ys is sorted (`ordered`). Assume ys is a sublist of xs (`subsequence`). Then

$$ys \text{ is a sublist of } \text{insertionSort}(xs).$$

Result 85 (ordered insert contains input). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Let a be a natural number. Then

$$xs \text{ is a sublist of } \text{insert}(a, xs).$$

Result 86 (euclidean coprime reduction). [Mathlib]
Let a, b and k be natural numbers. Then

$$a \text{ and } b + k\,a \text{ are coprime if and only if } a \text{ and } b \text{ are coprime.}$$

Result 87 (euclidean coprime subtraction). [Mathlib]
Let a and b be natural numbers. Assume $a \leq b$ (`ordered`). Then

$$b - a \text{ and } a \text{ are coprime if and only if } b \text{ and } a \text{ are coprime.}$$

Result 88 (coprime lcm product). [Mathlib]
Let a and b be natural numbers. Assume a and b are coprime (`cop`). Then

$$\text{lcm}(a, b) = a\,b.$$

Result 89 (common divisor of coprimes). [Mathlib]
Let a, b and k be natural numbers. Assume a and b are coprime (`cop`). Assume k divides a (`divides_a`). Assume k divides b (`divides_b`). Then

$$k = 1.$$

Result 90 (coprime product divisibility). [Mathlib]
Let a, b and k be natural numbers. Assume k and b are coprime (`cop`). Then

$$k \text{ divides } a\,b \text{ if and only if } k \text{ divides } a.$$

Result 91 (insertion sort certificate). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{sorting certificate}(xs, \text{insertionSort}(xs)).$$

Adapted from the cited source.

Result 92 (merge sort certificate). [Mathlib]
Let xs be a list with entries in $\mathbb{N}$. Then

$$\text{sorting certificate}(xs, \text{mergeSort}(xs)).$$

Adapted from the cited source.

Result 93 (certified sort unchanged). [Mathlib]
Let xs and ys be lists with entries in $\mathbb{N}$. Assume sorting certificate(xs, ys) (`certificate`). Then

$$\text{insertionSort}(ys) = ys.$$

Adapted from the cited source.

6 References

Mathlib The Mathlib Community. The Lean Mathematical Library. Version 4.24.0. Revision f897ebcf72cd. Source

Stewart 2001 G. W. Stewart. Matrix Algorithms, Volume II: Eigensystems. 2001. SIAM.